

EXAMPLE 1. The LCM of 30, 250, 490 is

- a. 46750 b. 36750
c. 26750 d. None of these

Sol. b. Here, $30 = 2 \times 3 \times 5$
 $250 = 5 \times 5 \times 2 \times 5 = 2 \times 5^3$
 and $490 = 7 \times 7 \times 2 \times 5 = 2 \times 5 \times 7^2$
 $\therefore$ LCM of 30, 250 and 490 $= 2 \times 5^3 \times 7^2 \times 3 = 36750$

2. **Division Method** Write the given number in a row and divide them with the common prime divisor. On division. Write the quotient in each case below the number. If any number is not divisible by the respective divisor, then write it as such in the next row. Keep on dividing the quotients until you get 1. Multiply all the divisors to get the required LCM.

EXAMPLE 2. What is the LCM of 120, 144, 160 and 180.

- a. 1450 b. 1620 c. 1440 d. 1380

Sol. c.

2	120,	144,	160,	180
2	60,	72,	80,	90
2	30,	36,	40,	45
3	15,	18,	20,	45
3	5,	3,	10,	15
5	5,	1,	10,	5
2	1,	1,	2,	1
	1,	1,	1	1

$\therefore$ LCM of 120, 144, 160 and 180 $= 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 2$
 $= 1440$

IMPORTANT POINTS

- The least number which when divided by x, y and z leaving the remainders a, b and c , respectively is given by $[\text{LCM of } (x, y, z) - p]$, where $p = (x - a) = (y - b) = (z - c)$.
- The least number which when divided by x, y and z leaving the same remainder R in each case is given by $[\text{LCM of } (x, y, z) + R]$.

EXAMPLE 3. What is the least number which when divided by 42, 72 and 84 leaves the remainders 25, 55 and 67, respectively?

- a. 521 b. 512 c. 504 d. 487

Sol. d. Here, difference $= (42 - 25) = (72 - 55) = (84 - 67) = 17$
 Now, LCM (42, 72, 84) = 504
 $\therefore$ Required number $= 504 - 17 = 487$

EXAMPLE 4. Find the least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 but when divided by 9, leaves no remainder.

- a. 1620 b. 1683
c. 1635 d. 1672

Sol. b. LCM (5, 6, 7, 8) = 840.

Here, $R = 3 \Rightarrow$ Number is of the form $840k + 3$.

Least value of k for which $(840k + 3)$ when divided by 9 leaves no remainder is 2.

$\therefore$ Required number $= 840 \times 2 + 3 = 1683$

Highest Common Factor (HCF)

- Common Factor** A common factor of two or more numbers is a number which divides each of them exactly. e.g. 2 is common factor of 2, 10, 20.
- Highest Common Factor** The Highest Common Factor (HCF) of two or more numbers is the largest number that divides all the given numbers exactly. It is also known as Greatest Common Divisor (GCD). HCF is always a factor of LCM. e.g. HCF of the numbers 18 and 24 is 6.

Methods of Finding HCF

- HCF by Prime Factorization** Write the given number as product of prime factors and then find the product of least powers of common prime factors. This product is the required HCF of given numbers.

EXAMPLE 5. The HCF of 65, 75 and 105 is

- a. 4 b. 5 c. 6 d. 8

Sol. b. Here, $65 = 13 \times 5$, $75 = 5 \times 5 \times 3$
 and $105 = 7 \times 3 \times 5$
 $\therefore$ HCF of 65, 75 and 105 = 5

- HCF by Division Method** Suppose we have to find the HCF of two given numbers, divide the large number by the smaller one. Now, you will get the remainder. Divide the divisor by the remainder. Repeat this process until no remainder is left, the last divisor used in this process is the desired greatest common divisor i.e. HCF. In order to find the HCF of three numbers, then, HCF of [(HCF of any two) and (third number)] gives the HCF of given three numbers.

EXAMPLE 6. The HCF of 204, 1190 and 1445 is

- a. 85 b. 15 c. 17 d. 75

Sol. c. Here, $1190 \div 1445$ (1

$$\begin{array}{r}
 1190 \\
 255 \overline{) 1190} \quad (4 \\
 \underline{1020} \\
 170 \quad 255 \quad (1 \\
 \underline{170} \\
 85 \overline{) 170} \quad (2 \\
 \underline{170} \\
 \hline
 \times
 \end{array}$$

So, HCF of 1190 and 1445 is 85.

$$\begin{array}{r}
 \text{Now,} \quad 85) 204 (2 \\
 \underline{170} \\
 34) 85 (2 \\
 \underline{68} \\
 17) 34 (2 \\
 \underline{34} \\
 \times
 \end{array}$$

$\therefore$ HCF of 85 and 204 is 17.
Hence, HCF of 204, 1190 and 1145 is 17.

IMPORTANT POINTS

1. For integers x , y and z , if $\text{HCF}(x, y) = 1$ and $\text{HCF}(x, z) = 1$, then $\text{HCF}(x, y, z)$ is always 1.
2. The greatest number that will divide x , y and z leaving remainders a , b and c , respectively is given by HCF of $(x - a)$, $(y - b)$, $(z - c)$.
3. The greatest number that will divide x , y and z leaving the same remainder in each case is given by $[\text{HCF of } |x - y|, |y - z|, |z - x|]$

EXAMPLE 7. Find the greatest number which will divide 400, 435 and 541 leaving 9, 10 and 14 as remainders respectively.

- a. 19 b. 17 c. 13 d. 9 **2014 I**

Sol. b Required number = HCF of $(400 - 9, 435 - 10, 541 - 14)$
= HCF of $(391, 425, 527) = 17$

EXAMPLE 8. For any integers ' a ' and ' b ' with $\text{HCF}(a, b) = 1$, what is $\text{HCF}(a + b, a - b)$ equal to?

- a. It is always 1 b. It is always 2 **2014 I**
c. Either 1 or 2 d. None of these

Sol. c. Put arbitrary values of a and b .

Illustration 1 Let $a = 9$ and $b = 8$.

$$\therefore \text{HCF}(8 + 9, 9 - 8) \Rightarrow \text{HCF}(17, 1) = 1$$

Illustration 2 Let $a = 23$ and $b = 17$.

$$\therefore \text{HCF}(17 + 23, 23 - 17) \Rightarrow \text{HCF}(40, 6) = 2$$

Hence, $\text{HCF}(a + b, a - b)$ can either be 1 or 2.

How to Calculate LCM and HCF of Fractions

The LCM and HCF of fractions can be obtained from the following formula

1. $\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$
2. $\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$

EXAMPLE 9. The HCF of $\frac{14}{3}, \frac{21}{9}, \frac{7}{15}$ is

- a. $\frac{7}{45}$ b. $3\frac{2}{5}$ c. $\frac{7}{30}$ d. None of these

Sol. a. HCF of numerators i.e. 14, 21 and 7 is 7 and LCM of denominators i.e. 3, 9 and 15 is 45. So, HCF of given fractions = $\frac{7}{45}$.

Relation between LCM and HCF of Two Numbers

Product of two numbers = (Their HCF) $\times$ (Their LCM)

EXAMPLE 10. The LCM of two numbers is 90 times their HCF. The sum of LCM and HCF is 1456. If one of the numbers is 160, then what is the other number?

2014 II

- a. 120 b. 136 c. 144 d. 184

Sol. c. Let the HCF of two numbers be x .
LCM of two numbers be $90x$.

According to the question,

$$\text{LCM} + \text{HCF} = 1456$$

$$\Rightarrow 90x + x = 1456$$

$$\Rightarrow 91x = 1456$$

$$\Rightarrow x = 16$$

$$\therefore \text{HCF of two numbers} = 16$$

$$\text{and LCM of two numbers} = 90 \times 16 = 1440$$

We know that,

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$\Rightarrow 1440 \times 16 = 160 \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{1440 \times 16}{160} = 144$$

EXAMPLE 11. What is the greatest number that divides 13850 and 17030 and leaves a remainder 17?

- a. 477 b. 159 c. 107 d. 87 **2012 II**

Sol. b. Required number = HCF of $(13850 - 17, 17030 - 17)$
= HCF of $(13833, 17013) = 159$

EXAMPLE 12. There are three drums with 1653 litre, 2261 litre and 2527 litre of petrol. The greatest possible size of the measuring vessel with which we can measure the petrol of any drum while every time the vessel must be completely filled is

- a. 31 b. 27 c. 19 d. 41

Sol. c. The maximum capacity of the vessel = HCF of 1653, 2261 and 2527 = 19

EXAMPLE 13. John, Kate and Smith at same time, same point and in same direction to run around a circular in 150 seconds. Find after what time will they meet again?

- a. 30 min b. 25 min c. 20 min d. 15 min

Sol. b. LCM of 250, 300 and 150 = 1500 sec
= 25 min.

Hence, John, Kate and Smith meet after 25 min.

> PRACTICE EXERCISE

- If $x = 2^3 \times 3^2 \times 5^4$ and $y = 2^2 \times 3^2 \times 5 \times 7$, then HCF of x and y is
(a) 180 (b) 360 (c) 540 (d) 35
- LCM of $2^3 \times 3 \times 5$ and $2^4 \times 5 \times 7$ is
(a) $2^{12} \times 3 \times 5^2 \times 7$ (b) $2^4 \times 5 \times 7 \times 9$
(c) $2^4 \times 3 \times 5 \times 7$ (d) $2^3 \times 3 \times 5 \times 7$
- LCM of $\frac{4}{5}$, $\frac{3}{10}$ and $\frac{7}{15}$ is
(a) $8\frac{2}{3}$ (b) $\frac{8}{15}$ (c) 20 (d) $16\frac{4}{5}$
- What is the HCF of $\frac{3}{2}$, $\frac{9}{7}$ and $\frac{15}{14}$?
(a) $\frac{3}{7}$ (b) $\frac{3}{14}$ (c) $\frac{3}{2}$ (d) 3
- The least number divisible by 12, 15, 20 and is a perfect square is
(a) 900 (b) 400 (c) 36 (d) 256
- The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 is
(a) 423 (b) 843 (c) 1683 (d) 2523
- The HCF of two numbers is $\frac{1}{5}$ th of their LCM. If the product of the two numbers is 720, then the HCF of the numbers is
(a) 13 (b) 12 (c) 14 (d) 18
- The LCM of two numbers is 39780 and their ratio is 13 : 15. Then, the numbers are
(a) 273, 315 (b) 2652, 3060
(c) 516, 685 (d) None of these
- If the highest common factor of two positive integers is 24, then their least common multiple cannot be
(a) 72 (b) 216 (c) 372 (d) 600
- If the HCF of three numbers 144, x and 192 is 12, then the number x cannot be
(a) 180 (b) 84 (c) 60 (d) 48
- Consider those numbers between 300 and 400 such that when each number is divided by 6, 9 and 12, it leaves 4 as remainder in each case. What is the sum of the numbers?
(a) 692 (b) 764 (c) 1080 (d) 1092
- What is the smallest positive integer which when divided by 4, 5, 8 and 9 leaves remainder 3, 4, 7 and 8, respectively?
(a) 119 (b) 319 (c) 359 (d) 719
- What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$?
(a) ab (b) a^2b^2 (c) a^2b^3 (d) a^3b^2
- If a number is exactly divisible by 11 and 13, which of the following types the number must be?
(a) Divisible by (11+13) (b) Divisible by (13-11)
(c) Divisible by (11×13) (d) Divisible by (13÷11)
- What is the sum of the digits of the least number which when divided by 52, leaves 33 as remainder, when divided by 78 leaves 59 and when divided by 117, leaves 98 as remainder?
(a) 17 (b) 18 (c) 19 (d) 21
- For any integer n , what is the HCF of integers $m = 2n + 1$ and $k = 9n + 4$?
(a) 3 (b) 1 (c) 2 (d) 4
- For any three natural numbers a, b and c , if HCF $\left(\frac{a}{c}, \frac{b}{c}\right)$ is
(a) a/c (b) b/c (c) c (d) Always 1
- Raj, Rachit and Asha begin to tag around a circular stadium. They complete their revolutions in 42 s, 56 s and 63 s, respectively. After how many seconds will they be together at the starting point?
(a) 366 (b) 252 (c) 504 (d) 605
- Find the side of the largest possible square slabs which can be paved on the floor of a room 2m 50cm long and 1m 50cm broad. Also, find the number of such slabs to pave the floor.
(a) 25,20 (b) 30,15 (c) 50,15 (d) 55,10
- Four bells begin to toll together and toll, respectively at intervals of 5, 6, 8 and 12 s. How many times will they toll together in an hour excluding the one at the start?
(a) 10 (b) 19 (c) 13 (d) 9
- 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. What is the minimum number of rows in which the above trees may be planted?
(a) 3 (b) 15 (c) 17 (d) 20
- A person has four iron bars whose lengths are 24 m, 36 m, 48 m and 72 m, respectively. This person wants to cut pieces of same length from each of four bars. What is the least number of total pieces, if he is to cut without any wastage?
(a) 10 (b) 15 (c) 20 (d) 25

- 23.** For two natural numbers m and n , let g_{mn} denote the greatest common factor of m and n . Consider the following in respect of three natural numbers k, m and n .

I. $g_{m(nk)} = g_{(mn)k}$ II. $g_{mn}g_{nk} = g_{mk}$

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 24.** Consider the following in respect of integers a and b
- I. $\text{HCF}(a, b) = \text{HCF}(a + b, b)$
II. $\text{HCF}(a, b) = \text{HCF}(a, b - a)$ for $b > a$
- Which of the above statement(s) is/are correct?
- (a) Only I (b) Only II
(c) Both I and II (d) None of these

PREVIOUS YEARS' QUESTIONS

- 25.** The HCF and LCM of two natural numbers are 12 and 72, respectively. What is the difference between the two numbers, if one of the numbers is 24? ✎ 2012 I
- (a) 12 (b) 18 (c) 21 (d) 24
- 26.** The sum of two numbers is 232 and their HCF is 29. What is the number of such pairs of numbers satisfying the above condition? ✎ 2012 I
- (a) One (b) Two (c) Four (d) None of these
- 27.** The product of HCF and LCM of 18 and 15 is ✎ 2012 II
- (a) 120 (b) 150 (c) 175 (d) 270
- 28.** Three planets revolve round the Sun once in 200, 250 and 300 days, respectively in their own orbits. When do they all come relatively to the same position at a certain point of time in their orbits? ✎ 2012 II
- (a) After 3000 days (b) After 2000 days
(c) After 1500 days (d) After 1200 days
- 29.** The LCM of two numbers is 2376 while their HCF is 33. If one of the numbers is 297, then the other number is ✎ 2013 I
- (a) 216 (b) 264 (c) 642 (d) 792
- 30.** The HCF of two numbers is 98 and their LCM is 2352. The sum of the numbers may be ✎ 2013 II
- (a) 1372 (b) 1398 (c) 1426 (d) 1484
- 31.** If for integers a, b and c , $\text{HCF}(a, b) = 1$ and $\text{HCF}(a, c) = 1$, then which one of the following is correct? ✎ 2013 II
- (a) $\text{HCF}(a, bc) = 1$ (b) $\text{HCF}(a, bc) = a$
(c) $\text{HCF}(a, bc) = b$ (d) None of these
- 32.** What is the number of integral solutions of the equations $\text{HCF}(a, b) = 5$ and $a + b = 65$? ✎ 2014 I
- (a) Less than 65 (b) Infinitely many
(c) Exactly one (d) None of these
- 33.** For any integer n , HCF of $(22n + 7, 33n + 10)$ is equal to ✎ 2014 I
- (a) n (b) 1 (c) 11 (d) None of these
- 34.** In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooter hit the target once in every 5 s, 6 s, 7 s and 8 s, respectively. If all of them hit their target at 9 : 00 am, when will they hit their target together again? ✎ 2014 I
- (a) 9 : 04 am (b) 9 : 08 am
(c) 9 : 14 am (d) None of these
- 35.** If a and b be positive integers, then HCF of $\left(\frac{a}{\text{HCF}(a, b)}, \frac{b}{\text{HCF}(a, b)}\right)$ equal to? ✎ 2014 I
- (a) a (b) b (c) 1 (d) $\frac{a}{\text{HCF}(a, b)}$
- 36.** The HCF of two natural numbers m and n is 24 and their product is 552. How many sets of values of m and n are possible? ✎ 2014 II
- (a) 1 (b) 2 (c) 4
(d) No set of m and n is possible satisfying the given conditions
- 37.** The LCM of two integers is 1237. What is their HCF? ✎ 2014 II
- (a) 37 (b) 19
(c) 1 (d) Cannot be determined
- 38.** There are 48 cricket balls, 72 hockey balls and 84 tennis balls and they have to be arranged in several rows in such a way that every row contains the same number of balls of one type. What is the minimum number of rows required for this to happen? ✎ 2014 II
- (a) 12 (b) 16 (c) 17 (d) 19
- 39.** Consider all positive two digit numbers each of which when divided by 7 leaves a remainder 3. What is their sum? ✎ 2015 II
- (a) 661 (b) 666 (c) 676 (d) 777
- 40.** What is the sum of digits of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11, respectively, as the remainders? ✎ 2015 II
- (a) 5 (b) 6 (c) 7 (d) 8
- 41.** The LCM of two numbers is 12 times their HCF. The sum of HCF and LCM is 403. If one of the numbers is 93, then the other number is ✎ 2015 II
- (a) 124 (b) 128 (c) 134 (d) 138
- 42.** Consider the following in respect of natural numbers a, b and c ✎ 2016 I
- I. $\text{LCM}(ab, ac) = a \text{ LCM}(b, c)$
II. $\text{HCF}(ab, ac) = a \text{ HCF}(b, c)$
III. $\text{HCF}(a, b) < \text{LCM}(a, b)$
IV. $\text{HCF}(a, b)$ divides $\text{LCM}(a, b)$.
- Which of the above statement(s) is/are correct?
- (a) III and II (b) III and IV
(c) I, II and IV (d) All of these

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42																	

HINTS AND SOLUTIONS

1. (a) Given, $x = 2^3 \times 3^2 \times 5^4$
and $y = 2^2 \times 3^2 \times 5 \times 7$
 $\therefore \text{HCF} = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$

2. (c) Here, say $a = 2^3 \times 3 \times 5$ and

$$b = 2^4 \times 5 \times 7, \text{ then}$$

$$\text{LCM} = 2^4 \times 3 \times 5 \times 7$$

3. (d) Here, LCM

$$= \frac{\text{LCM of } 4, 3, 7}{\text{HCF of } 5, 10, 15} = \frac{84}{5} = 16\frac{4}{5}$$

4. (b) $\text{HCF} \left(\frac{3}{2}, \frac{9}{7}, \frac{15}{14} \right) = \frac{\text{HCF}(3, 9, 15)}{\text{LCM}(2, 7, 14)}$

$$= \frac{3}{14}$$

5. (a)
- | | | | |
|---|-----|-----|----|
| 2 | 12, | 15, | 20 |
| 2 | 6, | 15, | 10 |
| 3 | 3, | 15, | 5 |
| 5 | 1, | 5, | 5 |
| | 1, | 1, | 1 |

LCM of 12, 15 and 20 = $2 \times 2 \times 3 \times 5$

$\therefore$ Required perfect square

$$= 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 900$$

6. (b) To find out the least number, firstly we find out the LCM of given numbers.

$$\therefore \text{LCM}(5, 6, 7, 8)$$

$$= \text{LCM}(5, 2 \times 3, 7, 2^3) = 840$$

$$\therefore \text{Required number} = 840 + 3 = 843$$

7. (b) Let $\text{LCM} = 5x$, then $\text{HCF} = x$
Now, product of numbers = 720

$$\text{So, } 5x \times x = 720$$

$$\Rightarrow 5x^2 = 720 \Rightarrow x = 12$$

8. (b) The numbers are $13x$ and $15x$.

So, x is the HCF. Now,

$$\text{HCF} \times \text{LCM} = \text{Product of numbers}$$

$$x \times 39780 = 13x \times 15x$$

$$\Rightarrow x \times 39780 = 13 \times 15 \times x^2$$

$$\Rightarrow x = \frac{39780}{13 \times 15} = 204.$$

$\therefore$ Numbers are $13 \times 204 = 2652$ and $15 \times 204 = 3060$

9. (c) In the given options, only 372 is not divisible by 24. Therefore, LCM of numbers cannot be 372.

10. (d) Here, we know that

$$144 = 12 \times 2 \times 2 \times 3$$

$$\text{and } 192 = 12 \times 2 \times 2 \times 2 \times 2$$

By taking option (d), $48 = 12 \times 2 \times 2$

Hence, the value of x will not be 48 otherwise the HCF of given numbers becomes 48.

11. (a) LCM of 6, 9 and 12 = 36
So, number is the form of $36p + 4$.

Since, the required numbers are between 300 and 400.

$$\therefore p = 9 \text{ and } 10$$

$$\therefore \text{Required sum} = 328 + 364 = 692$$

12. (c) Here,

$$4 - 3 = 5 - 4 = 8 - 7 = 9 - 8 = 1$$

$$\text{Now, } 4 = 2 \times 2, 5 = 5$$

$$8 = 2 \times 2 \times 2, 9 = 3 \times 3$$

$$\therefore \text{LCM} = 5 \times 2 \times 2 \times 2 \times 3 \times 3 = 360$$

$$\text{Required number} = 360 - 1 = 359$$

13. (b) $a^2 b^4 + 2a^2 b^2 = a^2 b^2 (b^2 + 2) \dots (i)$

$$\text{and } (ab)^7 - 4a^2 b^9 = a^7 b^7 - 4a^2 b^9$$

$$= a^2 b^2 (a^5 b^5 - 4b^7) \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\text{HCF} = a^2 b^2$$

14. (c) LCM of 11 and 13 will be (11×13) . Hence, if a number is exactly divisible by 11 and 13, then the same number must be exactly divisible by their LCM i.e. (11×13) .

15. (a) Here, $52 - 33 = 78 - 59$

$$= 117 - 98 = 19$$

$$\text{Now, } 52 = 13 \times 2 \times 2$$

$$\Rightarrow 78 = 13 \times 2 \times 3$$

$$\Rightarrow 117 = 13 \times 3 \times 3$$

$$\therefore \text{LCM} = 13 \times 2 \times 2 \times 3 \times 3 = 468$$

$$\therefore \text{Required number} = 468 - 19 = 449$$

Hence, the sum of the digits is 17.

16. (b) Since, $m = 2n + 1$ is an odd integer, so its factors may be 1 or 3 and $k = 9n + 4$ its factors may be 1, 2 and 4.

Hence, HCF of (m, k) is 1.

17. (d) It is always 1

Illustrations Let $a = 21$ and $b = 35$

Then, $\text{HCF}(21, 35) = 7$

$$\therefore \text{HCF} \left(\frac{21}{7}, \frac{35}{7} \right) = \text{HCF}(3, 5) = 1$$

18. (c) Required time = LCM of 42, 56 and 63 s

2	42, 56, 63
3	21, 28, 63
7	7, 28, 21
	1, 4, 3

$\therefore$ Required time

$$= 2 \times 3 \times 7 \times 4 \times 3 = 504 \text{ s.}$$

19. (c) Side of largest possible square slab is the HCF of 250 cm and 150 cm

$$250 = 5 \times 5 \times 5 \times 2$$

$$150 = 5 \times 5 \times 3 \times 2$$

$\therefore$ HCF is 50. Then, number of slabs

$$= \frac{\text{Area of floor}}{\text{Area of slab}} = \frac{250 \times 150}{50 \times 50} = 15$$

20. (a) Here, LCM of 5, 6, 8 and 12 is 360, so the bells will toll after 360 s.

So, in an hour they will toll together

$$= \frac{60 \times 60}{360} = 10 \text{ times}$$

21. (c) The HCF of (21, 42, 56) is 7.

$\therefore$ The minimum number of rows

$$= \frac{21}{7} + \frac{42}{7} + \frac{56}{7}$$

$$= 3 + 6 + 8 = 17$$

22. (b) Here, $24 = 12 \times 2$, $36 = 12 \times 3$

$$48 = 12 \times 4, \quad 72 = 12 \times 6$$

$$\therefore \text{HCF}(24, 36, 48, 72) = 12$$

$$\text{Total piece} = \frac{24}{12} + \frac{36}{12} + \frac{48}{12} + \frac{72}{12}$$

$$= 2 + 3 + 4 + 6 = 15$$